The Wrong Unit of Uncertainty: Simultaneous Inference for Repeated Cross-National Surveys

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Abstract

Repeated cross-national surveys are used to make claims about whole trajectories of national response distributions, while uncertainty is computed for single wave contrasts. When country trajectories calibrate cross-country prediction, moreover, the calibration objects are themselves complex-sample estimates. We develop simultaneous inference that attaches uncertainty to both units. Within a country, one design-based band certifies a partially ordered family of trajectory claims, from a single wave pair to persistent decline across a preregistered core of the distribution, with no further correction. Across countries, a clustered conformal band gives finite-sample coverage for survey-estimated trajectories. We prove that correcting toward the latent trajectory is unidentified without design-noise information, and characterize when the correction is unnecessary, not reliably learnable at the available K, or feasible: infeasible across countries, active one unit down. In the European Social Survey a marginal reading flags twenty of thirty countries; six certify net decline over 2018–2024, and over 2002–2024 none certifies a persistent slide, a claim that only crisis-scale change reliably clears, while closed testing certifies at least six true span decliners among thirty-three. The same procedure cuts the certified WVS deconsolidation set severalfold, toward post-communist and Arab-Spring states.

1 Introduction

Comparative politics increasingly treats a national public as a *distribution* rather than a mean (DiMaggio et al., 1996). Trust in parliament in a country and year is a cumulative distribution function over a response scale, and repeated cross-national surveys, among them the European Social Survey (ESS; Kaminska and Lynn, 2017), the World Values Survey (WVS), and the AmericasBarometer (Cohen et al., 2021), generate claims about how those distributions evolve across waves. A second use, *transporting* trajectories (predicting one country's from the others'), draws on conformal prediction's finite-sample, distribution-free coverage (Vovk et al., 2005; Lei et al., 2018) and its shift relaxations (Tibshirani et al., 2019; Gibbs and Candès, 2021; Gibbs et al., 2024).

Two wrong units. One principle organizes this paper: *the stochastic unit of a guarantee must coincide with the unit of the scientific claim it certifies* (Robinson, 1950; Openshaw, 1984). Claims from repeated surveys violate it twice. Uncertainty is routinely reported for a local contrast, one wave pair at one threshold, while the *claim* it is taken to support ("democratic trust is declining") concerns a whole trajectory. The deconsolidation debate of Foa and Mounk (2016, 2017) and Valgarðsson et al. (2025) is the substantive case we re-examine, not a methodological culprit, since its own evidence is cohort comparisons. The second, subtler wrong unit is the calibration: the objects conformal transport calibrates on are themselves *estimated* from complex probability samples, and existing constructions plug them in as if exact (Dunn et al., 2023; Lee et al., 2026).

The two errors are distinct. Writing $F_{c,r}(t)$ for country c's latent CDF at threshold t in wave r and $\widetilde{F}_{c,r}(t)$ for its complex-survey estimate,

$$\widetilde{F}_{c,r}(t) - \mu_r(t) \;=\; \underbrace{\big(F_{c,r}(t) - \mu_r(t)\big)}_{\text{transport error}} \;+\; \underbrace{\big(\widetilde{F}_{c,r}(t) - F_{c,r}(t)\big)}_{\text{design error}}, \tag{1}$$

where μ_r is the transport center. A band calibrated on the observed $\widetilde{F}$ scores treats the sum as if it were the transport error alone. Its spread mixes cross-population variation with survey-design variation that is irrelevant to the latent target, so the band can be inefficient (under a scale family its target scale is inflated by the familiar $1/\sqrt{1-\rho^2}$). Yet a correction that subtracts the survey noise, itself estimated from a handful of countries, can make the band too narrow.

Contributions. *First, a procedure for the claim unit*: a design-based simultaneous band over a country's whole trajectory. From it a partially ordered family of claims, from a single wave pair up to persistent decline over a preregistered core of the distribution, is read off one coverage event (Section 3.1). *Second, an identification boundary for the calibration unit.* Without information on the design-noise law the latent transport-error law is non-identified, and no band built on the observed curves can uniformly improve on the plug-in conformal band beyond its discreteness slack while remaining valid (Theorem 1). *Third, the feasibility boundary of the correction, and a selector that respects it.* Even with the design-noise information supplied, the correction's reliability diagnostic obeys an algorithmic floor $D \geq \sqrt{2/(K-1)}$, so the deployed correction requires $K \geq 94$ exchangeable populations. The two largest surveys fail this bound in opposite ways, while one unit down, on regions within countries ($K \approx 230$–290), the same frozen selector activates (Section 6). The anchor band is finite-sample valid at every K, exact under its symmetric centers and carried into the released leave-one-out deployment by a $K/(K-1)$ inflation (Theorems 3 and 5). The clustered band itself is machinery shared with a companion paper on a non-survey domain (Park, 2026); new here is what the survey layer forces: the identification boundary, the floor, the safe selector, and the claim-family deployment. *Fourth, two reanalyses in which the procedures change the substantive picture* (Section 7). For European trust, net decline certifies in six of thirty countries (2018–2024) against the twenty a marginal reading flags, and over 2002–2024 no country certifies persistence while eight certify span erosion, all read off one band per country. Globally, a trajectory-persistence criterion cuts the Foa–Mounk "deconsolidation" set on the WVS trend file (1981–2022) by 2.6–6.5×, leaving a thirteen-country descriptive co-certified core; twelve pass a dependence-robust test of at least two true item declines. The geography remains post-communist and Arab-Spring-aftermath states.

2 Setup and the two targets

Objects. We observe K source countries, indexed $c = 1, \ldots, K$, treated as exchangeable units. Each country has a latent *response-distribution trajectory* $F_{c,r}(t)$, a CDF over a fixed grid of thresholds $t \in \{t_1, \ldots, t_T\}$ and survey rounds r. We say *response distribution* deliberately: reading a certified shift as an *attitude* shift also requires measurement invariance across waves (Section 8), though every claim below survives monotone recoding of the ordinal scale. We do not see $F_{c,r}$. We see a complex-survey estimate $\widetilde{F}_{c,r}(t) = F_{c,r}(t) + S_{c,r}(t)$, where $S_{c,r}$ is design (sampling) error with a design variance $v_{c,r}(t)^2$ that a stratified primary-sampling-unit (PSU) or replicate-weight bootstrap can estimate. Three curves carry the paper: the latent deviation from the transport center, $G_c = \{F_{c,r}(t) - \mu_r(t)\}_{r,t}$; the design error S_c; and their observable sum $\widetilde{G}_c = \{\widetilde{F}_{c,r}(t) - \mu_r(t)\}_{r,t} = G_c + S_c$, which is the decomposition (1). The transport score of a country is its worst-case standardized deviation over the trajectory,

$$R_c \;=\; \max_{r,t} \frac{|G_{c,r}(t)|}{\sigma(t)}, \qquad \widetilde{R}_c \;=\; \max_{r,t} \frac{|\widetilde{G}_{c,r}(t)|}{\sigma(t)},$$

with $\sigma(t)$ a fixed modulation profile (the deployed anchor takes $\sigma \equiv 1$, the unstudentized score). R_c is latent, $\widetilde{R}_c$ observed, and $\xi_c := \widetilde{R}_c - R_c$ is a difference of maxima, bounded by $|\xi_c| \le \max_{r,t} |S_{c,r}(t)|/\sigma(t)$; it is not additive noise, and nothing below treats it as centred.

Design noise is measured at the curve coordinate, where the additive model holds. Let $s^2_{\text{plug}}(t)$ be the between-country variance of $\widetilde{G}_c(t)$, $s^2_R(t)$ that of $G_c(t)$, and $\overline{v^2}(t)$ the mean design variance. For design error independent of the latent curve, $s^2_{\text{plug}}(t) = s^2_R(t) + \overline{v^2}(t)$, and the *design share*

$$\rho(t)^2 \;=\; \frac{\overline{v^2}(t)}{s^2_{\text{plug}}(t)} \;=\; \frac{\overline{v^2}(t)}{s^2_R(t) + \overline{v^2}(t)} \;\in\; [0, 1)$$

is the governing quantity; design noise is negligible as $\rho \to 0$ and dominant as $\rho \to 1$. The scalar $\widehat{\rho}$ that we report and gate on is a conservative aggregate of this profile, the ratio of the t-averaged lower confidence bound on $\overline{v^2}(t)$ to the t-averaged upper bound on $s^2_{\text{plug}}(t)$. It is a diagnostic, not a variance decomposition of the sup-score; the point estimate reported beside it drops the bounds, $\sqrt{\overline{\widehat{v}^2}}/\overline{s_{\text{plug}}}$.

Target and validity. We want a simultaneous band covering a new country's *latent* trajectory (target T2), $\Pr\left(F_{\text{new},r}(t) \in \widehat{B}(r,t)\ \forall r, t\right) \geq 1-\alpha$; the *survey-estimate* trajectory $\widetilde{F}_{\text{new}}$ is target T1. Ordinary clustered conformal on the observed scores is exact for T1 under exchangeability and a symmetric center (Theorem 3). Its latent coverage differs by an explicit gap that vanishes with the design-noise scale (Theorem 2(b); symmetry alone is not enough), and its spread carries design variation, a scale inflation by $1/\sqrt{1-\rho(t)^2}$ under a scale family, whose removal is a deconvolution. Table 1 collects the procedure–target–guarantee triples.

What country exchangeability assumes. The transport band's identifying assumption is a superpopulation postulate: participating countries' trajectory laws are symmetric draws over a set that is not sampled and enters rounds non-randomly. Three things temper it. The headline certifications of Section 7 do not use it. Under approximate exchangeability the Barber et al. (2023) bound controls the gap. And a leave-one-region-out audit finds near-nominal held-out coverage except for the post-communist East (Supplementary Material).

Procedure	Target	Key assumption	Guarantee
Clustered conformal band (anchor)	survey-estimated trajectory	country exchangeability	finite-sample, every K, under symmetric centers (Thm 3)
Same band, latent reading	latent trajectory	+ design-noise tail and anti-concentration	$1-\alpha-$gap, gap shrinking with the design-noise scale (Thm 2b)
Design-aware correction	latent trajectory	design law (A1)–(A3), $K \geq 94$	$1-\alpha-\varepsilon_{K,B}$; deployed level a frozen calibrated bound (Thm 4)
Certification band (§3.1)	within-country contrast surface	survey-design bootstrap	asymptotic simultaneous $1-\alpha$

Table 1: Which procedure guarantees what. The conservative envelope (omitted) shares the anchor's row, containing it by construction; the theory's finite-sample language does not transfer to the certification band.

The impossibility. The deconvolution is not identified from the contaminated curves alone. The result is stated at the curve level of (1), where the additive model *is* the survey sampling structure; the design noise may be arbitrarily heteroskedastic across thresholds.

Theorem 1 (Non-identification and necessity of design information)**.** *Let the observed deviation curves be $\widetilde{G}_c = G_c + S_c$ as in* (1)*, the latent curves exchangeable with law $\mathcal{L}_G$, the design-noise curves independent of them with law $\mathcal{S}$ in an admissible class that contains $S \equiv 0$ and is closed under one further nondegenerate symmetric convolution. (i) $\mathcal{L}_G$ is not identified from the observed law $\mathcal{L}_G * \mathcal{S}$. (ii) Any band whose radius is measurable in the observed curves alone and is valid for the latent target over the whole admissible class cannot uniformly improve on the plug-in conformal band: on the member $S \equiv 0$, which it cannot distinguish from the contaminated configurations, the plug-in radius is already the tightest valid one up to the conformal discreteness slack, which randomized smoothing exhausts. Uniform improvement requires shrinking the class of noise laws.*

The reading is constructive (proof in the Supplementary Material): a band both valid and narrower than the plug-in must supply the design-noise law, which is what Section 3's design bootstrap does.

Remark 1 (Beyond surveys). The identification problem is not survey-specific: it recurs wherever calibration objects carry a nonidentified additive error (model-based small-area outputs, learned or imputed features, meta-analytic effects). The K-floor is narrower, a property of the unbiased-variance estimator class (Proposition 2). A non-survey small-area simulation confirms the point (Supplementary Material).

Relation to existing work. Conformal prediction has met complex surveys at the unit level (Wieczorek, 2023) and *Political Analysis* once (Randahl et al., 2026); cluster-as-unit constructions assume clean cluster samples (Dunn et al., 2023; Lee et al., 2026), ours complex designs (Binder, 1983; Lumley, 2010). Noise-aware conformal methods assume a *known* noise model (Einbinder et al., 2024; Sesia et al., 2025; Singer et al., 2025); Theorem 1 says that premise fails for design noise, a deconvolution problem (Fan, 1991), and Proposition 2 that even a supplied law faces a finite-K floor.

3 Simultaneous inference for both units

The two wrong units get two procedures: within-country certification of a whole claim family, and cross-country transport with a correction that activates only where it is provably reliable.

3.1 Within-country certification: one band, a family of claims

Write $\theta_{a,b}(t) = F_b(t) - F_a(t)$ for the contrast between observed rounds $a < b$ at threshold t, positive where response mass has moved below t, a decline. Let $\mathcal{T}_0$ be the preregistered core of thresholds. Table 2 defines each claim as an event for this contrast surface. Persistent implies both net and any-pair, but net and any-pair are incomparable: a recovered dip certifies an adjacent pair without net decline, and a slow accumulation certifies a span while no adjacent pair clears the critical value. Nor are the claims separate tests: each claim, every sub-span, and the reverse direction is a function of one object, the country's surface of wave-pair contrasts, so all are certified on one coverage event.

Claim	Event A_j for the surface θ	Certified when
Pairwise $(a, a+1)$	$\theta_{a,a+1}(t) > 0 \ \ \forall t \in \mathcal{T}_0$	$\min_{t \in \mathcal{T}_0} \left[\widehat{\theta}_{a,a+1}(t) - c\,\widehat{s}_{a,a+1}(t)\right] > 0$
Any-pair	$\exists\, a :\ \theta_{a,a+1}(t) > 0 \ \ \forall t \in \mathcal{T}_0$	some adjacent pair certifies
Net (span)	$\theta_{1,R}(t) > 0 \ \ \forall t \in \mathcal{T}_0$	the first-to-last contrast certifies
Persistent	$\theta_{a,a+1}(t) > 0 \ \ \forall t \in \mathcal{T}_0,\ \forall a < R$	every adjacent pair certifies
Recovery (a, b)	$\theta_{a,b}(t) < 0 \ \ \forall t \in \mathcal{T}_0$	the reverse inequality, same c
Across countries	conjunction over countries	Bonferroni, or closed testing (§7)

Table 2: The claim family as events for one country's contrast surface $\theta = \{\theta_{a,b}(t)\}$ over R observed rounds. $\widehat{\theta}$ and $\widehat{s}$ are the design-bootstrap estimate and standard error of each contrast and c the single sup-t critical value shared by every entry and both directions, so every certification is read off the one event $\theta \in \widehat{B}$. Persistent implies net and any-pair; net and any-pair are incomparable.

Proposition 1 (Claim-family transfer)**.** *Let $\widehat{B}$ be a simultaneous $1-\alpha$ band for a country's contrast surface $\theta = \{\theta_{a,b}(t)\}$ over every ordered pair of observed rounds, and let $\{A_j\}$ be any family of claims, each a set of surfaces. Define the certified subfamily $J = \{j : \widehat{B} \subseteq A_j\}$. Then $\Pr\big(\theta \in A_j$ for every $j \in J\big) \geq 1-\alpha$, with no dependence on $|J|$: on $\{\theta \in \widehat{B}\}$, set inclusion delivers every certified claim at once.*

This is the post-hoc logic of Scheffé (1953), of closed testing (Marcus et al., 1976), and of the selection-free bounds of Goeman and Solari (2011); Katsevich and Ramdas (2020). The contribution is the deployment: trend claims are made claim by claim, whereas one sup-t band over the whole contrast family removes the per-claim corrections (no Bonferroni across a country's pairs, no budget for recoveries, no endpoint-selection cost) at the price of a wider critical value (Section 7). The persistent object is one functional band (Diquigiovanni et al., 2022), an estimand distinction rather than an error-rate one (Gelman et al., 2012; Benjamini and Hochberg, 1995). The band itself is a design-based sup-t band on wave-pair CDF differences from the stratified-PSU bootstrap (Rao and Wu, 1988; Wolter, 2007), an asymptotic guarantee to which the theory sections' finite-sample language does not transfer.

3.2 Cross-country transport and the safe adaptive band

Theorem 1 says the design bootstrap is the identifying input; the transport method turns it into a band by choosing, target-blind, among three constructions.

Three branches. Take calibration curves $\{\widetilde{G}_c\}_{c=1}^K$ with per-country design SD $\widehat{v}_c(t)$ from the stratified-PSU bootstrap. *(i) The clustered conformal band* takes the conformal quantile of the *unstudentized* scores $\max_t |\widetilde{G}_c(t)|$, exact for the survey-estimate target at any K under Theorem 3's symmetric centers. *(ii) Deconvolution* studentizes by the finite-K-guarded target scale $\widehat{s}_T^2(t) = \max\big(s_{\text{plug}}^2(t) - [\overline{\widehat{v}^2}(t) - z\ \text{SE}]_+,\ \text{floor}\big)$. The guard subtracts a one-sided margin below the mean design variance ($z = 1.645$ deployed), so that on the guard event the correction cannot over-shrink. Under (A1) the branch's target scale, though not its realized width, contracts by $\sqrt{1-\rho(t)^2}$ coordinate by coordinate. *(iii) The conservative band* inflates each score by the design SD, $\max_t(|\widetilde{G}_c(t)| + z\,\widehat{v}_c(t))$, which dominates the clustered score pointwise, so the conservative band *contains* the clustered band by construction.

Safe selector (Algorithm 1). A naive selector asks *can we deconvolve* and *do we need to*; the safe selector adds *can we do it safely at this K?* Deconvolution is used only when four target-blind gates pass, all computed from the calibration set. The *need* gate requires $\widehat{\rho}_{\text{LCB}} > \rho_0$, the aggregate design share against a frozen operational cutoff. The *reliability* gate requires $D = \max_t \text{SE}(\widehat{s}_T^2)/\widehat{s}_T^2 \le \tau_D$, a frozen bound on the finite-K remainder (Proposition 2). The *efficiency* gate requires a certified width gain over the conservative band, and the *stability* gate requires the guarded scale to stay above its floor. Otherwise the band is an anchor. All constants are frozen before validation (Supplementary Material). Theorem 5 shows that validity survives any selection between the anchors, the deconvolution branch being charged its own budget only where its gate can open at all ($K \ge 94$). Three levels differ: the *nominal* $1-\alpha$; the *theorem* level $1-\alpha-\varepsilon_{K,B}$ of Theorem 4, whose remainder is unobservable; and the *validated operational* level reported when the procedure deconvolves, $1-\alpha-\widehat{\delta}_{\text{UCB}}(D)$, a

Algorithm 1: Nominal-safe adaptive design-aware band

Input: calibration curves $\{\widetilde{G}_c\}$ (unit = country), design replicates $\{\widehat{v}_c\}$, transport center $\widehat{\mu}$ (leave-one-out mean in the unsurveyed-target default), level α; frozen gate constants and the K-only budget rule $\alpha_{\text{dec}}(K)$

set $(\alpha_{\text{anchor}}, \alpha_{\text{dec}})$ by the frozen budget rule (full α to the anchors when gate B is infeasible at this K);

compute diagnostics s, $\widehat{s}_T$, $\widehat{\rho}_{\text{LCB}}, D, \widehat{\delta}_{\text{UCB}}(D)$; the nested unstudentized anchor radii $r_{\text{clu}}, r_{\text{con}}$ at level α_{anchor}; and r_{dec} at level α_{dec};

if $\widehat{\rho}_{LCB} \leq \rho_0$ **then**

  branch $\leftarrow$ clustered;

else if $\widehat{\delta}_{UCB}(D) \leq \delta_{\max}$ ***and*** *stable* ***and*** $\widehat{Gain} - \Delta_g > g_{\min}$ **then**

  branch $\leftarrow$ deconvolution;

else

  branch $\leftarrow$ conservative;

if branch $\neq$ deconvolution **and** the center is leave-one-out (the unsurveyed-target default) **then**

  $r_{\text{branch}} \leftarrow \frac{K}{K-1} r_{\text{branch}}$; `// exactness for the deployed center (Section 4)`

return $\widehat{\mu} \pm r_{\text{branch}}$, branch, and diagnostics;

bound frozen before validation and checked by it. The released implementation returns the band, the branch it selected, and the level that band carries.

Computational assistance. Between June and September 2026, the author used Claude Opus versions 4.8–5 via Claude.ai for limited assistance with figure-generation code and selected analysis-code implementation. The author designed the study and methods, wrote the manuscript, ran and reviewed all analyses, and independently verified every AI-assisted code path and reported result. The author takes full responsibility.

4 Theory

Table 1 maps each procedure to the target it covers. Proofs are in the Supplementary Material, Section S1.

Theorem 2 (Oracle validity)**.** *Suppose the countries are exchangeable and the design law is known. (a) The clustered band is finite-sample valid for the survey-estimate target,* $\Pr(\widetilde{F}_{new} \in$

$\widehat{B}) \geq 1-\alpha$*, using exchangeability only. (b) For the latent target the same band satisfies, for every* $u > 0$, $\Pr(F_{new} \in \widehat{B}) \geq 1-\alpha-\Pr(\xi < -u) - \Pr(\widetilde{R} \in (\widehat{q}-u, \widehat{q}])$*. (c) Under the scale-family condition (A1), the oracle deconvolution band with the true target scale is finite-sample valid for the latent target and exact at the attainable conformal rank, and its target scale is a factor* $\sqrt{1-\rho(t)^2}$ *of the observed scale at each coordinate. The realized width carries, in addition, the ratio of the two conformal quantiles, which tends to one as* $\rho \to 0$*. Some such shape restriction is necessary (Supplementary Material).*

The gap in (b), a noise-tail term plus a local anti-concentration term, vanishes with the design-noise scale (order $\sigma_\xi^{2/3}$ under variance-only tails, σ_ξ up to logarithms under sub-Gaussian noise, σ_ξ a scale for ξ) but not by symmetry alone (a bimodal counterexample is in the Supplementary Material). Hence we estimate and report the design share rather than assume it away.

Theorem 3 (Exact finite-K validity of the base band)**.** *Let the base band use the unstudentized cluster score of the* observed *curves,* $\widetilde{R}_c = \max_t |\widetilde{G}_c(t)|$ *(t ranging over the stacked round-by-threshold grid; the latent* R_c *of Section 2 is unobserved), with a center satisfying the symmetric-construction condition (the Supplementary Material states the four admissible cases). If the countries are exchangeable, then for every K the band* $\widehat{B} = center \pm \widehat{q}$*, with* $\widehat{q}$ *the* $\lceil(1-\alpha)(K+1)\rceil$*-th order statistic of* $\{\widetilde{R}_c\}_{c\leq K}$ *(and* $\widehat{q}=\infty$ *when that rank exceeds K, i.e. when* $K < \alpha^{-1}-1$*), satisfies* $\Pr(\widetilde{F}_{new} \in \widehat{B}) \geq \lceil(1-\alpha)(K+1)\rceil/(K+1) \geq 1-\alpha$*, distribution-free, with equality at the stated rank level when the scores are almost surely distinct. The guarantee holds at every K; the band is informative only for* $K \geq \alpha^{-1}-1$*.*

Exactness at every K is bought by excluding two asymmetries, in-sample modulation and grand-mean centers that contain their own unit, at a width cost of at most 5% on the near-homoskedastic real curves.

Theorem 4 (Estimated-law validity)**.** *Assume a common scale family indexed by the design variance (A1), anti-concentration of the sup-score law near its upper quantile (A2), and* *bounded curves with a size-B design bootstrap whose ideal variance is the design variance (A3), stated in full in the Supplementary Material with the bootstrap's two layers separated. Then (i) with oracle scales the deconvolution band is finite-sample valid for the latent target at every K and exact at the attainable conformal rank, and (ii) with estimated scales it satisfies* $\Pr(F_{new} \in \widehat{B}) \geq 1 - \alpha - \varepsilon_{K,B}$ *with* $\varepsilon_{K,B} = O\big(\sqrt{\log(KT)/\min(K,B)}\,\big)$.

By construction the finite-K-guarded scale $\widehat{s}_T^2$ of Section 3 subtracts a one-sided margin below the mean design variance, so on the guard event (Supplementary Material) the design-variance-subtraction component of the scale error can only widen the band. The sampling error in s^2_{plug} stays two-sided and is absorbed into $\varepsilon_{K,B}$. (A1) matches the *distributions* of standardized observed and latent curves, more than equal covariance profiles give; Gaussian families satisfy it. The survey-scale guarantee does not rest on it: where it is doubtful the selector abstains.

The two clauses below cover different targets: the survey-estimated trajectory below the floor, the latent one above it.

Theorem 5 (Safe-adaptive finite-K validity)**.** *Let the anchors use Theorem 3's centers (or the leave-one-out center with its $K/(K-1)$ inflation, Propositions S1 and S3). For the first clause assume the countries' observed trajectories exchangeable; for the second assume (A1)–(A3). Deploy Algorithm 1 with the unstudentized clustered and conservative branches, scores* $\max_t |\widetilde{G}_c(t)|$ *and* $\max_t(|\widetilde{G}_c(t)| + z\widehat{v}_c(t))$*, at level* α_{anchor}*, and the deconvolution branch at* α_{dec}*. Then, with no condition on the selection rule,*

$$\Pr\left(\widetilde{F}_{new,r}(t) \in \widehat{B}(r,t)\ \forall r,t\right) \geq 1-\alpha \quad (K < 94),$$

$$\Pr\left(F_{new,r}(t) \in \widehat{B}(r,t)\ \forall r,t\right) \geq 1-\alpha-\varepsilon_{K,B} \quad (K \geq 94),$$

the first clause finite-sample, the second carrying Theorem 4's remainder.

Proof sketch. Below the floor gate B cannot open (Proposition 2(b) is a deterministic identity of the frozen diagnostic), so the selector only ever chooses between the two anchors.

Their scores are ordered pointwise, $|\widetilde{G}_c(t)| + z\widehat{v}_c(t) \geq |\widetilde{G}_c(t)|$, so at a common level the conservative order statistic is at least the clustered one and the bands are nested, $\widehat{B}_{\text{clu}} \subseteq \widehat{B}_{\text{con}}$. A miss of whichever band is selected implies a miss of $\widehat{B}_{\text{clu}}$, which Theorem 3 bounds by α. This is the first clause; it conditions on no selection event (Gupta et al., 2022; Yang and Kuchibhotla, 2024). At $K \geq 94$ the budget splits, $\alpha = \alpha_{\text{anchor}} + \alpha_{\text{dec}}$, by a frozen rule of K alone. Under (A1) each observed calibration score dominates its latent counterpart, since $\sqrt{s_R^2 + v_c^2} \geq s_R$ pointwise, so the anchor quantile stochastically dominates the latent-score order statistic, and the anchors are valid for the *latent* target at α_{anchor} (anchor-domination lemma, Supplementary Material). With the deployed leave-one-out center they pay a same-order centering term (Proposition S3), absorbed into $\varepsilon_{K,B}$. The non-nested deconvolution branch is charged its own budget by a union bound, $\Pr(\text{miss}) \leq \Pr(\text{miss}, \widehat{J} \neq \text{dec}) + \Pr(\text{miss}, \widehat{J} = \text{dec}) \leq \alpha_{\text{anchor}} + \alpha_{\text{dec}} + \varepsilon_{K,B}$, the last term from Theorem 4(ii) for the always-deconvolve band, its own centering term priced the same way (Supplementary Material). No term conditions on $\widehat{J}$.

A scope note. The unsurveyed-target deployment uses a leave-one-out center, outside Theorem 3's symmetric constructions, but inflating the radius by $K/(K-1)$ restores finite-sample validity at every K, distribution-free (Proposition S1). The released default ships that inflation at a width cost of $1/(K-1)$, 3.5% at the ESS's $K \approx 30$. The exact split-fold alternative is infeasible at survey scale: halving the calibration set below $\alpha^{-1}-1$ units makes the conformal radius infinite.

Efficiency. Under (A1) the deconvolution band reduces continuously to the clustered band as $\rho \to 0$, with oracle scale factor $\sqrt{1-\rho(t)^2} = 1 - \frac{1}{2}\rho(t)^2 + O(\rho(t)^4)$ at each coordinate; the conservative band contains the clustered one pointwise; and selection cannot inflate miscoverage (Theorem 5).

5 Simulation

Simulation establishes two facts: the wrong unit collapses coverage, and the safe selector meets its guarantees on unseen data.

5.1 The wrong unit collapses coverage

We simulate $K = 30$ countries and $T = 6$ thresholds over trajectories of length L rounds, with errors correlated across rounds and thresholds, and ask each band for *trajectory* coverage (Figure 1). All three bands are unstudentized and finite-sample, so only the score's unit varies. A band that is marginally valid at the round level covers the trajectory 81.7% of the time at $L = 2$ and 49.8% at $L = 8$; per threshold the figures are 38.2% and 3.5%. Only the country-trajectory band holds its nominal 90% at every length, because the trajectory is a single exchangeable unit. The obvious rejoinder, correcting the wrong-unit bands for multiplicity, fails at survey scale. A Bonferroni-corrected per-round band needs a conformal quantile at level α/L, which exists only for $K \geq L/\alpha - 1$ countries (79 at $L = 8$; 479 for the threshold-level band), so at $K = 30$ its radius is infinite for every $L \geq 4$. Where it does exist ($K = 100$), it is 4–26% wider than the trajectory band for comparable or higher coverage (90–94% against 89–91%). The choice is therefore not whether to correct but which unit carries the simultaneous statement at the K surveys have. The multiplicity-corrected baselines are a post-preregistration robustness benchmark.

Severity: what each claim can detect. At ESS-like design noise, 80% power at the *net* claim needs a per-pair decline of 0.02–0.03 CDF points. The *persistent* claim needs 0.06–0.08, a crisis-scale change, and its power *falls* as the trajectory lengthens. Injecting known declines into the real ESS pairs puts the thresholds at ≈ 0.033 (net) and ≈ 0.075 (persistent), with realized size 0.001–0.007 against a nominal 0.10. The procedure is markedly conservative, so non-certification is weak evidence of absence at every claim. Persistent is the strongest rung, not the sensitive one: gradual cumulative erosion is the net and span claims' object

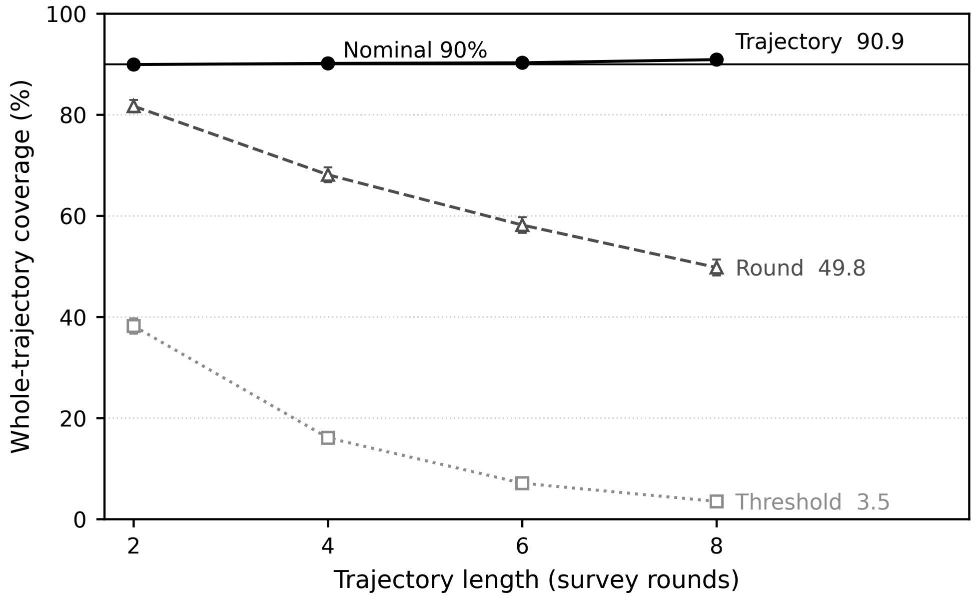

Figure 1: Coverage of a whole country trajectory, by the unit the score is attached to (4,000 replicates, nominal 90%; bars are ±2 Monte Carlo SEs). Only the trajectory-unit band holds its level as the trajectory lengthens.

(Section 7).

Validation. The frozen procedure was then evaluated on a sealed set of six unseen data-generating families, with the script hash recorded before first execution. All 120 cells were compatible with the target coverage (worst 0.878), with zero activations below $K = 94$, as the floor requires. The three-layer validation history (including a first holdout demoted to development after a scorer audit) is in the Supplementary Material.

6 Evidence from the ESS and the AmericasBarometer

The regime characterization. At the cross-national unit the selector reduces to the clustered conformal band. This is the selector's own low-ρ gate at work, not an assumption. Across both surveys and every estimand examined, $\widehat{\rho} \leq 0.22$ on the AmericasBarometer and at most 0.29 in the ESS subgroup scan, well below the frozen need cutoff $\rho_0 = 0.47$. Every $\widehat{\rho}$ we report is a downward-biased *floor* in expectation: the PSU bootstrap resamples

m-of-m without Rao–Wu–Yue rescaling (Rao et al., 1992). The rescaled rerun leaves the regime intact (maxima 0.28 and 0.22, zero activations, every certification count unchanged), and subsampling the AmericasBarometer to smaller designs never fires the selector either (Supplementary Material).

Why the deconvolution is unreachable.

Proposition 2 (Finite-K feasibility of the deployed correction)**.** *The deployed selector activates deconvolution only if a* need *gate* $\widehat{\rho}_{\mathrm{LCB}} > \rho_0$ *and a* reliability *gate* $D \le \tau_D$ *both hold, with* $D = \max_t \widehat{\mathrm{SE}}(\widehat{s}_T^2)/\widehat{s}_T^2$ *and the frozen* $\widehat{\mathrm{SE}}(\widehat{s}_T^2) = \sqrt{2s_{\mathrm{plug}}^4/(K-1) + (\mathrm{SD}_K(\widehat{v}^2)/\sqrt{K})^2}$*. Both gates are bounded by* algorithmic identities of the frozen procedure. *(a)* $\rho(t)^2 = \overline{v^2}(t)/s_{\mathrm{plug}}^2(t) \in [0,1)$ *coordinate-wise, and* $\widehat{\rho}_{\mathrm{LCB}}$ *is its conservative aggregate: the need gate fires only when design noise is an appreciable fraction of the between-population spread,* ρ_0 *being a frozen operational threshold. (b) Since* $\widehat{s}_T \le s_{\mathrm{plug}}$ *by construction,* $D \ge \sqrt{2/(K-1)}$*, so the reliability gate requires* $K \ge 1 + 2/\tau_D^2$*, i.e.* $K \ge 94$ *at the frozen* $\tau_D = 0.147$*. (The Supplementary Material separates the deterministic statement from its kurtosis-dependent interpretation.)*

The two requirements pull apart on real surveys (Figure 2). On the ESS the noisiest subpopulations raise $\widehat{\rho}$ only to 0.29, while $K \le 33$ leaves $D \ge 0.25$; neither gate opens (42 cells, zero activations). The WVS is the mirror image. Its full item-coverage sets ($K = 95$–105) make the reliability gate feasible and the certification samples ($K = 59$–77) do not, and either way $\widehat{\rho}_{\mathrm{LCB}} \le 0.10$ fails the need gate. The survey large enough to estimate the deconvolution has almost no design noise to remove; the one with real clustered noise has far too few countries. Nor is the floor an artifact of our thresholds. At the i.i.d. Gaussian member, any *unbiased* variance estimator from K populations has relative standard error at least $\sqrt{2/(K-1)}$ (Lehmann–Scheffé; Supplementary Material), so any correction that demands relative reliability τ from such an estimate needs $K \ge 1 + 2/\tau^2$. The frozen τ_D is one instantiation, 94 its intercept, $\sqrt{2/(K-1)}$ the law for this estimator class. With the proved oracle scale-factor gain $1-\sqrt{1-\rho^2}$, this splits the (K,ρ) plane into regimes where the correction is *unnecessary*, *unlearnable* (within the unbiased-variance class), or *feasible*; this paper's datasets occupy the three corners.

Where the correction does live: small areas, on this same survey. The barrier is about the unit, and the ESS contains a finer one. Take a region's departure from its own *national* distribution, $D_g = F_g - F_{\text{nation}}$, calibrate the transport band across regions, and take the design SD of the difference from one stratified-PSU bootstrap of the whole country (Supplementary Material). There the design share is not marginal: $\widehat{\rho}_{\text{LCB}}$ runs 0.26–0.52 and clears ρ_0 in eight of twenty-three *unit-rounds* (minimum unit size $\times$ round). Both gates open once $K \gtrsim 230$, at units of forty to sixty respondents in rounds 9 and 10, $K = 228$–287. There the *frozen* selector activates the design-aware branch for the first time on real survey data, a feasibility demonstration rather than a validation of latent-target coverage, since the latent departure is unobserved. The band is 21–27% narrower than the conservative envelope at its validated operational level 0.88, with certified width-gain lower bounds of 0.16–0.22 across twelve independent bootstrap streams. The activation survives the Rao–Wu–Yue rescaled bootstrap, though the activating cells fall from four to one (Supplementary Material).

The regions nest in twenty-seven countries, nine each, so the objection this paper would itself raise is that the effective unit is the country; holding out whole countries is the diagnostic. Calibrating on the other twenty-six countries' regions leaves the plug-in anchor at 0.89–0.90 against its 0.90 target in all twelve cells, within 0.015 of leave-one-region-out calibration. Coverage *conditional* on the country does not transfer; the worst held-out countries sit at 0.53–0.63. The guarantee is marginal over regions and presumes their exchangeability, which shared national dependence can violate; the diagnostic bounds that concern without removing it. In the activating cells the selected band covers 0.93–0.98 against its 0.88 on the observed departure; latent-target validity rests on (A1). Two scope conditions: reaching $K \gtrsim 230$ pools regions coded at different NUTS levels: among the fourteen countries that

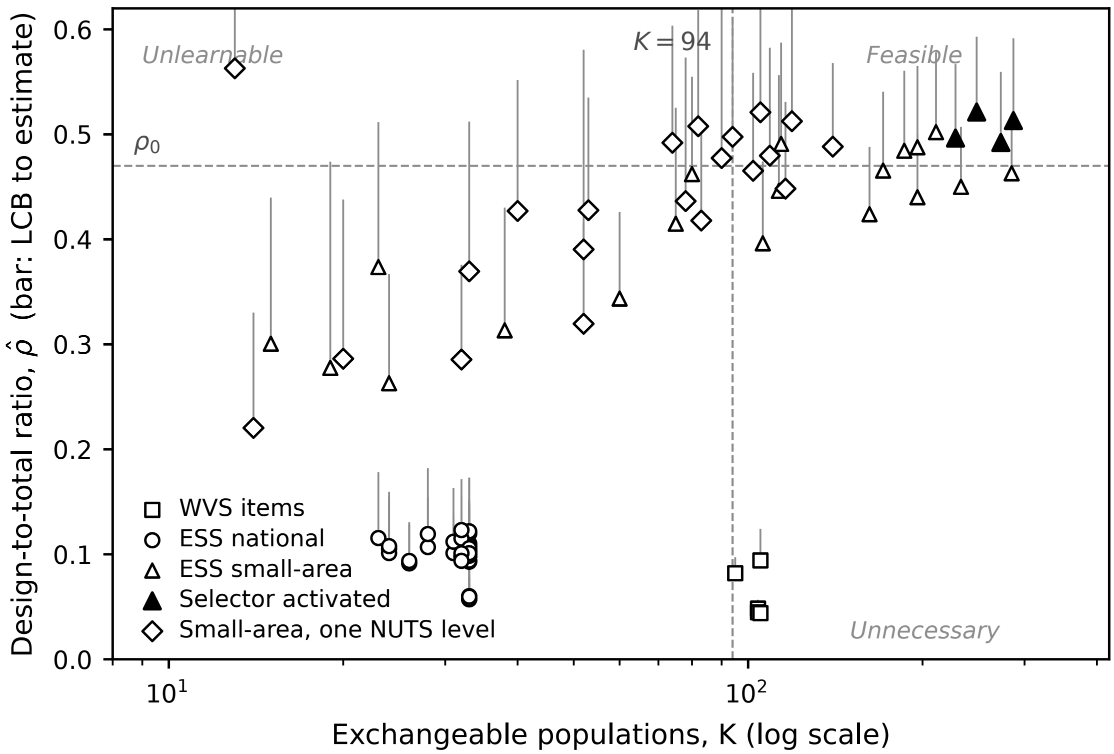

Figure 2: The feasibility frontier in the $(K, \hat{\rho})$ plane. Each marker is a real-data cell at its lower confidence bound, with a bar to the point estimate; filled markers are the four cells where the frozen selector activated. Dashed rules are the frozen boundaries, the need cutoff ρ_0 and the reliability floor $K = 94$.

share one level the need gate opens but $K \leq 140$ keeps the reliability gate shut. And round 11 never activates (Supplementary Material). The correction is therefore reachable in principle, unreachable across countries, and operationally feasible at the small-area scale, where Theorem 1's remark placed it.

7 Political reanalysis: how much decline survives simultaneous inference?

We analyze trust in parliament (`trstprl`) and satisfaction with democracy (`stfdem`) in ESS rounds 9–11 (2018–2024), the window whose integrated files ship PSU/stratum identifiers, then extend every trajectory to the full record (rounds 1–11, 2002–2024), asking, claim by claim, how many countries survive. The comparative literature has grown skeptical of a continent-wide trust decline (Norris, 2017; Voeten, 2017; Wuttke et al., 2022); we add a formal procedure. Dominant trend estimates (Valgarðsson et al., 2025) rest on latent-trait models with no design-based uncertainty layer (Tai et al., 2024), and latent-mood estimation (Claassen, 2019, 2020a; Kołczyńska and Bürkner, 2026) pools countries into model-based credible intervals; our bands give model-free coverage at a stated claim, a complement. Certification levels here are design-bootstrap simultaneous levels, asymptotic in the within-country samples (Table 1).

The counts collapse across claims. For `trstprl` over rounds 9–11, a marginal pairwise reading flags 20 of 30 countries. An any-pair design-aware band drops this to 12. Net first-to-last decline holds in 6 (Austria, Belgium, Estonia, the United Kingdom, Greece, the Netherlands). The *persistent, design-aware simultaneous* band over the preregistered low-trust core certifies one, Greece, which also survives Bonferroni across countries. *Greece fielded only rounds 10 and 11 here*, so its trajectory is a single pair and the top claims coincide for it (as for Czechia and Israel, uncertified). Among the 27 two-pair countries none certifies persistence. The United Kingdom, Belgium, and the Netherlands pass the *plug-in* persistent claim and are demoted by the design-aware layer alone. The headline is two numbers: net decline in six of thirty, and *zero* of the twenty-seven countries observed across multiple pairs certifying persistence. The severity analysis (Section 5) explains why: the persistent claim has 80% power only against crisis-scale declines, so its non-certifications are weak evidence of absence. For `stfdem` the pattern repeats (23 marginal, 14 any-pair, 8 net, one persistent: Greece, again one pair, failing Bonferroni there). The literature's Greek legitimacy collapse (Armingeon and Guthmann, 2014; Torcal, 2014) concerns 2009–2015; the certified pair is 2020/21→2023/24.

The long window: the secular question, asked properly. We next extend every trajectory to its full record (rounds 1–11; median six pairs, up to ten; rounds 1–8 weights-only, the sample-design-file merge changing no net or persistent count, Supplementary Material). Zero of the thirty-three countries with three or more usable rounds certify persistent decline over the preregistered core across their full record, plug-in included. Reversing the inequality certifies *recoveries*, and across the 1,173 ordered contrasts of the trust series certified movement is directionally balanced (193 declining against 212 rising). Inside the same band, at the same α, twenty-three of thirty-three countries certify at least one decline and at least one recovery on each outcome (twenty on both); the two directions share one critical value, so no within-country correction is needed. The count across countries carries no familywise control by itself; the closed-testing bound below supplies it. The certified record therefore argues against describing European trust as a uniformly monotone slide, on the positive evidence of coexisting certified declines and recoveries rather than on the persistent rung's non-certifications.

Under Proposition 1 (one band, every ordered contrast, both directions, $\alpha = 0.10$ jointly), net erosion over the whole span certifies in eight countries (Cyprus, Spain, the United Kingdom, Greece, Hungary, Israel, Italy, Ukraine; Figure 3, Table 3). Against the per-claim construction the joint statement costs France at the span claim ($9 \to 8$) and six countries at the any-pair claim ($28 \to 22$); the persistent zero holds under both constructions (Supplementary Material).

The across-country count itself can now carry a guarantee. Inverting each joint band over its level yields a per-country certification p-value, and closed testing across the thirty-three (Goeman and Solari, 2011) then certifies, with 90% simultaneous confidence, that *at least six* truly declined over their span on each outcome, the six nameable at no further cost (Supplementary Material). The Simes local tests behind that bound need the countries' p-values independent or positively dependent. That is a design assumption (independent sampling mechanisms given the finite populations), not a property of separate bootstrap streams. Bonferroni local tests, which assume nothing about dependence, return the same six on both outcomes, as does the rounds 1–8 design-file upgrade (Supplementary Material).

Table 3: Certified net decline in parliamentary trust, 2002–2024, read off each country's joint band (every ordered contrast, both directions, $\alpha = 0.10$). *Decline* $\geq$: simultaneous lower bound over the preregistered core, in CDF points. *Contrasts*: ordered round pairs certified declining, of all admissible; their share is the endpoint-free erosion index. The certified set is stable under weights-only design-effect inflation up to roughly 1.3; above that Spain, the marginal member, drops out (Supplementary Material).

country	rounds	decline $\geq$	contrasts	share
Ukraine	2–11	25.9	7/15	0.47
Cyprus	3–11	6.2	13/21	0.62
United Kingdom	1–11	6.1	11/55	0.20
Greece	1–11	3.1	8/15	0.53
Israel	1–11	2.6	10/28	0.36
Italy	1–11	1.6	5/15	0.33
Hungary	1–11	1.1	9/55	0.16
Spain	1–11	0.6	32/55	0.58

The ladder of units thus closes at the top: threshold, wave, trajectory, country, count.

Under-detection shows in the ledger of contrasts (Table 3): Hungary and the United Kingdom certify their 22-year spans on a fifth or less of their ordered contrasts, erosion a reading built from adjacent waves would mostly miss. Because every endpoint choice is certified at once, the band also delivers the *share of a country's ordered contrasts that certify a decline*. That share is a lower bound, comparable within a country but not across, because the shared critical value increases strongly with record length while the denominator grows quadratically; so we do not rank on it. Share and magnitude differ: Spain certifies 32 of 55 contrasts yet clears its span by only 0.6 CDF points. Ukraine's certified span decline of at least 25 points runs from the post-Orange-Revolution wave of 2005 to a wartime sample: a certified statement about those two waves.

The reclassified countries are weak-signal, higher-noise cases, *inconclusive under the design-aware band* rather than false positives.

Two confounds this window cannot exclude. Round-10 fieldwork coincided with the pandemic's rally-round-the-flag movements, so "persistent decline through this window" is a claim made against a rally; and nine countries switched to self-completion at round 10, confounding wave-pair shifts with mode effects that no design bootstrap absorbs. A mode audit bounds the latter: the Greek pair is mode-constant, and net counts are unchanged when the switchers' confounded pairs are excluded (Supplementary Material, with a preregistered age-group analysis finding no pooling artifact).

A second, global target: Foa–Mounk deconsolidation. We preregistered a reanalysis of the "democratic deconsolidation" thesis (Foa and Mounk, 2016, 2017) on the WVS trend file (1981–2022, 59–77 countries with ≥ 2 waves), recoding the five-item battery so that deconsolidation is a persistent decline over each scale's preregistered core. The harmonized trend file provides no design metadata, so this is an external substantive stress test of the thesis, not a second validation of the method; the ESS carries the design-based validity story. The verdict is calibration rather than denial, but the surviving phenomenon is not the one the thesis was about. A *marginal wave-pair reading of the same data*, constructed to instantiate the bottom rung, flags 36–64 countries per item, against 6–17 (8–24%) at the persistent rung. That 2.6–6.5× ratio compounds two changes. Holding the inference layer fixed, the rung alone accounts for 1.7–4.8× (plug-in) or 1.9–4.8× (design-aware); propagating design uncertainty at a fixed rung accounts for the remainder. Three caveats. The trend file carries no PSU or stratum identifiers, so the weights-only bootstrap omits within-PSU dependence (variance-inflating) and stratification (variance-reducing). Where the former dominates, as is usual, certified counts are *inflated*: at variance ×1.5 and ×2.0 the descriptive core holds at 13 and 12 countries and the gap widens to 3.0–8.3×, so the reported gap is then a lower bound. The stronger partial-conjunction set described next moves from 12 to 10 and 9, respectively. The per-item lists carry no across-country familywise control. And the certification sets sit below the $K \geq 94$ bound, so every WVS band uses the full-α anchors of Theorem 5.

The co-certified core: where the surviving deconsolidation lives. Aggregating per-item certifications yields a positive characterization (Figure 4): of 38 countries certified on ≥ 1 item, 13 form the descriptive multi-item core on ≥ 2. Inverting each persistent band

and applying an arbitrary-cross-item-dependence partial-conjunction test (Benjamini and Heller, 2008) upgrades that reading: 12 countries each pass a 90% test of at least two true item declines; only Trinidad and Tobago drops ($p_{\mathrm{PC}} = 0.144$). Each guarantee is within-country; closing all 78 country p-values yields a lower bound of one requiring no cross-country dependence assumption, naming Rwanda. The twelve-country set is concentrated in post-communist and Arab-Spring states, not the mature democracies the thesis concerned, and it formalizes, with per-country error control, the decline-from-euphoric-baselines pattern described qualitatively (Mishler and Rose, 1997; Inglehart and Catterberg, 2003). Three qualifications. Five of the twelve partial-conjunction members are electoral autocracies in the V-Dem classification, where falling stated support reflects autocratic legitimation rather than deconsolidation in the consolidated-democracy sense. The West enters the core twice, through Finland and Switzerland on the strongman/army pair, which we flag as a probable administration artifact; no consolidated Western democracy certifies on support for a democratic system, consistent with Wuttke et al. (2022). And the geography is partly window-contingent: recertifying on fieldwork ending by 2017 keeps eight of the thirteen but drops all three Arab-Spring cases, so the post-communist component is window-stable and the Arab-Spring component rests on WVS wave 7. Certified attitudes do not forecast institutions: descriptive-core membership predicts subsequent V-Dem electoral-democracy decline no better than the marginal flag (Fisher $p = 0.89$; Ouattara and van der Meer, 2023; van der Meer and Janssen, 2025), so certification is descriptive error control, not a forecast.

8 Discussion

Uncertainty in claims about repeated cross-national surveys is usually computed for the wrong unit, twice. We have given a multiplicity-free simultaneous band for the claim unit and a hard boundary for the calibration unit: the correction is non-identified without the design-noise law (Theorem 1), and unreachable for the deployed procedure at cross-national scale even with it (Proposition 2). At the national unit the method reduces to clustered population conformal, and the reported design share certifies that the simple band is the right one. The deconvolution regime needs many exchangeable units and comparable design noise, never both across countries but reachable one unit down (Section 6).

Limitations. The machinery prices sampling error only: measurement non-invariance (wording changes, house effects, mode switches) produces the very shifts the bands certify, so every certification is conditional on measurement comparability. The deployed m-of-m PSU bootstrap (Rao et al., 1992; Wolter, 2007) biases design variance downward; the Rao–Wu–Yue rescaled rerun leaves every certification count and cross-national gate unchanged, while the small-area activation set is bootstrap-sensitive in the disclosed direction (Supplementary Material). Transport-band exchangeability is a superpopulation postulate over a non-sampled, panel-attriting country set, and holds worse for the post-communist bloc. And without design metadata (the WVS trend file) only weights-only replication is possible.

The general lesson. Attaching uncertainty to the right unit changes substantive conclusions: no persistent decline certified among thirty-three ESS countries over the full record, yet span erosion certified in eight, mostly invisible to adjacent-wave readings; a certified deconsolidation set several-fold smaller, concentrated outside the consolidated West. The pattern is not confined to surveys: wherever prediction is calibrated on estimated objects (Remark 1), the calibration targets carry their own error, and the claim unit is often coarser than the trend asserted. The right response is to state the strongest object the data support, correct estimation noise where it is identified, and abstain where it is not, rather than to widen every band reflexively.

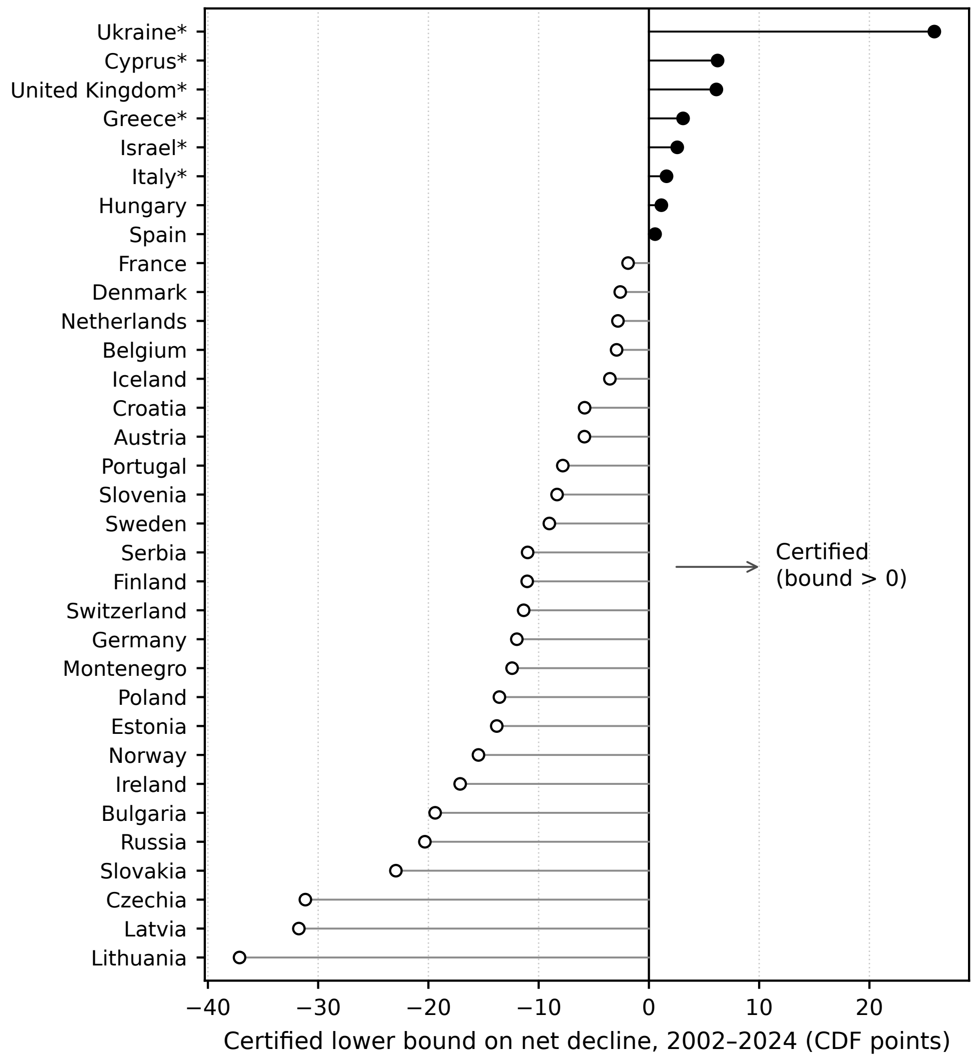

Figure 3: Each country's simultaneous lower bound on the net (first-to-last) decline in parliamentary trust, 2002–2024, read off its joint band at $\alpha = 0.10$. Filled markers: the eight countries whose bound exceeds zero; asterisks: the six named by the closed-testing prevalence bound. Negative bounds say how far a country sits from certifying, not that it improved.

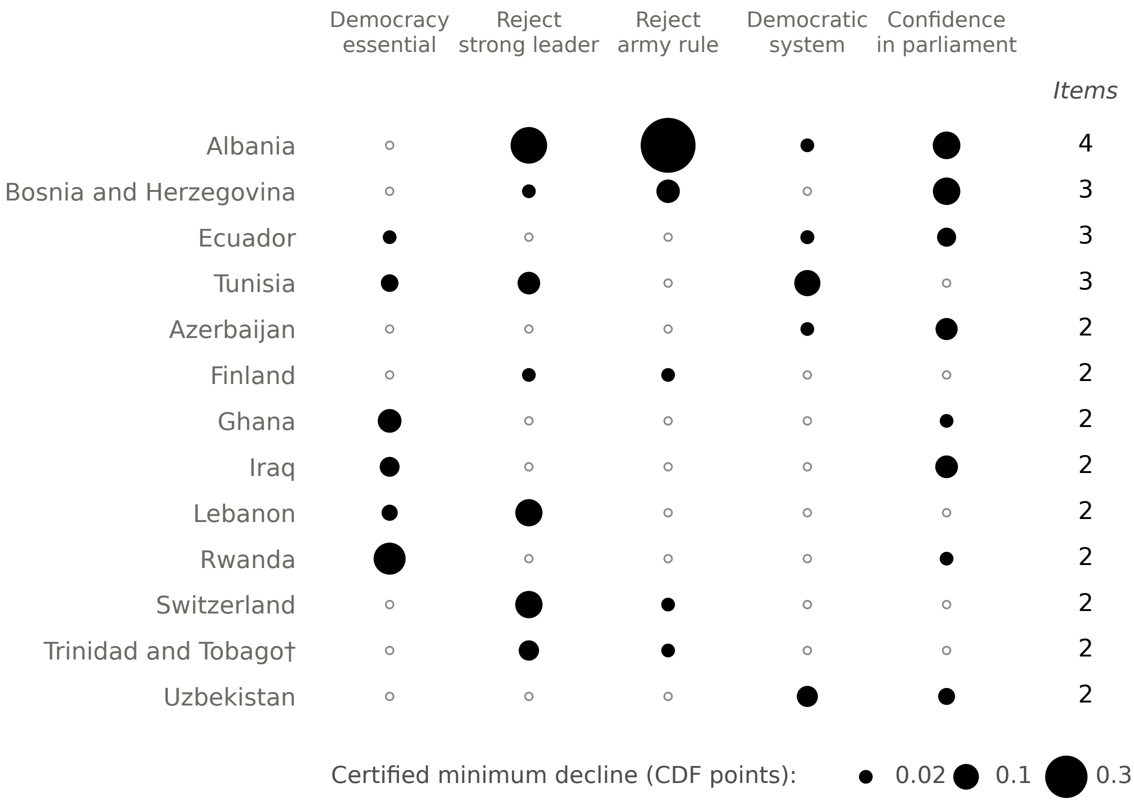

Figure 4: The thirteen-country descriptive co-certified core, by evidence strength (WVS 1981–2022; $K = 59$–77 per item). Twelve pass the arbitrary-cross-item-dependence ≥ 2-item partial-conjunction test, which does not say which two; the dagger marks Trinidad and Tobago, which does not pass. Circle area is the persistent band's simultaneous lower bound on the per-pair decline, in CDF points; open circles mark items that do not certify. The full thirty-eight-country matrix is in the Supplementary Material.

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Data Availability Statement

Replication code for this article is available at `https://github.com/kunhailP/design-aware-conformal`; the package will be deposited in the Political Analysis Dataverse with a version tag upon conditional acceptance. Simulation, theory, and benchmark results are deterministic under fixed seeds with no external data, and each theorem is paired with an automated contract test checking its computational implications. The survey microdata (European Social Survey ERIC, 2024; Haerpfer et al., 2024; LAPOP Lab, 2023) are licensed to their providers and cannot be redistributed; all code that runs on them is public, and the replication package documents the exact file editions, variables, retrieval steps, and sha256 checksums, alongside the public V-Dem and Claassen inputs (V-Dem Institute, 2025; Claassen, 2020b). Three headline analyses have been verified to reproduce the committed result tables bit-identically from the raw licensed files in independent environments. The editor has been notified of the restricted access at submission.

Competing Interests

The author declares none.

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